

Ontology Quality Report

Generated automatically from `generate_ontology_report.py`
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Validation Results

File	Syntax	HermiT	Pellet	OOPS
A.owl	✓	✓	✓	✓
B.owl	✓	✗	✗	✓
C.owl	✓	✓	✓	✓
D.owl	✓	✓	✓	✓

Extracted Details

File	Classes	Obj Props	Data Props	Ann Props	Triples	Max Depth	Avg Depth	Max Branch	Leaves	RR	AR	IR	Axiom Div.	Name Strict	Name Usc.	Name Bad	Label Cov.	Comment Cov.
A.owl	24	34	12	0	411	1	0.21	2	21	0.8718	0.5	0.2083	7	1.0	0.0	0.0	1.0	1.0
B.owl	74	69	46	0	1045	4	0.77	8	55	0.6509	0.6216	0.5	6	0.6085	0.3915	0.0	0.5291	0.3862
C.owl	19	36	14	0	386	5	1.74	4	12	0.7826	0.7368	0.5263	6	1.0	0.0	0.0	1.0	1.0
D.owl	22	30	12	0	340	1	0.41	4	18	0.7692	0.5455	0.4091	6	1.0	0.0	0.0	1.0	1.0

Metric Definitions

Structural Ratios (OntoQA Framework)

- RR (Relationship Richness)** = $\frac{|\text{ObjectProperties}|}{(|\text{subClassOf axioms}| + |\text{ObjectProperties}|)}$. Ratio of non-taxonomic to all relations. Higher values indicate a more interconnected graph rather than a flat taxonomy.
- AR (Attribute Richness)** = $\frac{|\text{DatatypeProperties}|}{|\text{Classes}|}$. Average number of data properties per class. Higher values indicate richer per-instance data modelling.
- IR (Inheritance Richness)** = $\frac{|\text{subClassOf triples}|}{|\text{all named classes}|}$. Average number of inheritance edges per class (including leaf classes). Distinct from Avg Branching which only considers parent classes.

Axiom Complexity

- **Axiom Diversity Score** (0–10): Count of distinct advanced OWL constructs present in the ontology, out of: `owl:someValuesFrom`, `owl:allValuesFrom`, `owl:disjointWith`, `owl:inverseOf`, `owl:equivalentClass`, `owl:unionOf`, `owl:intersectionOf`, `owl:hasValue`, and cardinality restrictions (`owl:cardinality`, `owl:minCardinality`, `owl:maxCardinality`, counted as one). A score of 0 means the ontology uses only declarations and `rdfs:subClassOf`; a score of 10 means all construct types are present.

Lexical & Annotation Quality

- **Naming: Strict CamelCase** (0.0–1.0): Fraction of named entities matching strict Semantic Web conventions — UpperCamelCase for classes (`^[A-Z][A-Za-z0-9]*$`), lowerCamelCase for properties (`^[a-z][A-Za-z0-9]*$`). No underscores.
- **Naming: Underscore Style** (0.0–1.0): Fraction that follow camelCase with underscores (e.g. `Cl_Invoice`), a common prefix-based variation that would pass if underscores are allowed but failed the strict check.
- **Naming: Non-conformant** (0.0–1.0): Fraction that match neither pattern. The three fractions sum to 1.0.
- **Label Coverage** (0.0–1.0): Fraction of named entities (classes + all property types) that have at least one `rdfs:label` triple. 1.0 = fully labelled.
- **Comment Coverage** (0.0–1.0): Fraction of named entities that have at least one `rdfs:comment` triple. 1.0 = fully documented.

Hierarchy Metrics

- **Max Depth**: Length of the longest `rdfs:subClassOf` chain from a root class (no parent) to a leaf class. Computed via BFS.
- **Avg Depth**: Mean depth across all named classes.
- **Max Branching**: Highest number of direct subclasses any single class has.
- **Avg Branching**: Mean number of direct subclasses per class, computed only over classes that have at least one child (excludes leaf classes).
- **Leaf Classes**: Count of classes with no direct subclasses.